

Theoretical Bounds for the Three Gaps in Learned Image Compression Quantization

Walker Kirkpatrick

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Abstract

Learned image compression (LIC) models achieve state-of-the-art rate-distortion performance but rely on several approximations that introduce gaps between the achieved rate and the information-theoretic limit. Pan et al. (2021) identified three such gaps: the discrete gap (continuous relaxation vs. hard quantization), the entropy estimation gap (learned model vs. true marginal), and the local smoothness gap (within-bin density variation). We derive tight upper bounds for each gap as a function of latent dimension, entropy model capacity, quantization step size, and training data size. We show that the total gap decomposes additively and identify which gap dominates in different operating regimes.

1 Introduction

Neural image codecs optimize the rate-distortion Lagrangian $L = \mathbb{E}[-\log P(k)] + \lambda \cdot d(x, \hat{x})$ via stochastic gradient descent. However, three key approximations introduce rate penalties:

1. **Discrete gap:** Replacing non-differentiable rounding with additive uniform noise during training.
2. **Entropy estimation gap:** Using a learned entropy model $P(k)$ instead of the true marginal $M(k)$.
3. **Local smoothness gap:** Assuming the density is constant within each quantization bin.

2 Main Results

Theorem 2.1 (Discrete Gap Bound). *For a latent representation of dimension d with effective standard deviation σ_{eff} and quantization step Δ :*

$$\Delta_{\text{discrete}} \leq d \cdot \log_2 \left(1 + \frac{\Delta}{2\sigma_{\text{eff}}} \right)$$

Theorem 2.2 (Entropy Estimation Gap Bound). *For an entropy model with N_{model} parameters trained on M samples:*

$$\Delta_{\text{entropy}} \leq \sqrt{\frac{d \cdot N_{\text{model}}}{M}} + d \cdot H(\Delta)$$

where $H(\Delta) = -\log_2(\Delta) \cdot \Delta^2/6$ is a bin-width correction.

Theorem 2.3 (Local Smoothness Gap Bound). *For a latent density with Lipschitz constant L :*

$$\Delta_{smoothness} \leq \frac{d}{2} \cdot \log_2 \left(1 + \frac{(L\Delta)^2}{12} \right)$$

Corollary 2.4 (Total Gap). *The total quantization gap satisfies:*

$$R_{achieved} - R(D) \leq \Delta_{discrete} + \Delta_{entropy} + \Delta_{smoothness}$$

3 Discussion

The bounds reveal that:

- The discrete gap dominates at coarse quantization (large Δ).
- The entropy estimation gap dominates when the model is overparameterized ($N_{\text{model}} \gg M$).
- The smoothness gap dominates for high-variability latents (large L).
- All gaps vanish as $\Delta \rightarrow 0$ (high-rate limit), recovering the asymptotic optimality of NTC.

References

- [1] S. Pan et al., “Three gaps for quantisation in learned image compression,” *CVPRW*, 2021.
- [2] J. Ballé et al., “Nonlinear transform coding,” *IEEE Trans. Inf. Theory*, 2021.